

R1

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April 1, 2026

1 New Code / Dev Stuff

1.1 Python Terminal Script for Filtering

Due to the fact that we need to filter multiple simulations using very similar methods, I decided to make an actual script instead of just using a notebook repeatably. The largest upside is that this can easily be run on a compute node, which makes filtering larger sims easier and faster. Having a single main filtering script also helps make sure any results are reproducible.

Takes a user input via the CLI (see below), filters for all **BH_Sol** systems, and returns the population file with said systems. Additionally, it returns a .csv of just the **BH_Sol** rows. While running, it provides feedback via the terminal stating how many **BH_MS** systems it found, as well as how many are the expected **S1:S2** configuration. If any are **S2:S1**, it prints a warning as well as the total count.

Need to further refine it, but there's also a default SBATCH script to help expedite the submission process.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
-o	output location	./results
-outputName	name for the output files ¹	BH_Sol_Systems
-overwrite	Whether to overwrite existing results, T/F bool. Defaults to F	True
--maxMass	Max mass used when filtering, defaults to $3M_{\odot}$	3.5
--maxPeriod	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

¹If a name is not specified it will use the original filename, appending a specifier to the front

2 New Research

2.1 Pop 1

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, ZAMS_oCE1_CC1_oRL02_CC2. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then into post-MS **unstable** RLO via $S1 \rightarrow S2$. This period decrease leads to common envelope, which then *greatly* decreases the orbital period. The envelope is then ejected, leaving a depleted stripped He core². The core then collapses in a couple of thousand years, leaving the system in a BH-MS state with low p_{orb} /separation.

These systems then evolve in a semi-peculiar way, where it is highly likely that there will be RLO from S2 onto S1, and because $S1 > S2$, this transfer is stable, leading to *very* prolonged stable RLO (seemingly on the scale of $> 10Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the initial period for all the systems is close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the initial masses of S1 and S2 appear to be quite constrained as well.

	orbital_period.i		S1_mass.i		S2_mass.i
count	6.000000	count	6.000000	count	6.000000
mean	4022.716197	mean	17.515761	mean	2.296788
std	151.522384	std	0.520269	std	0.320702
min	3873.118288	min	16.815111	min	1.797809
25%	3908.300983	25%	17.190036	25%	2.120753
50%	3978.305326	50%	17.510176	50%	2.353828
75%	4124.397101	75%	17.818155	75%	2.547210
max	4248.507680	max	18.254965	max	2.620529

2.2 Pop 2

I ran a second grid including an eccentricity distro (Ps and Qs), as well as slightly constraining the initial conditions ($12 < M_1 < 100, .1 < M_2 < 10$)³, hoping to confirm some of the init conditions.

	orbital_period.i		S1_mass.i		S2_mass.i
count	12.000000	count	12.000000	count	12.000000
mean	4136.066721	mean	2.240382	mean	17.357478
std	177.676729	std	0.412075	std	1.073496
min	3844.570082	min	1.551266	min	15.506937
25%	4024.214908	25%	1.903135	25%	16.714083
50%	4119.397535	50%	2.300024	50%	16.998229
75%	4273.345845	75%	2.579111	75%	18.162633
max	4370.857404	max	2.729828	max	19.435080

²I need to read a lot more about CE evo, but my intuition feels weird about the fact that just the envelope of *one* of the stars is ejected

³This is very constrained very early, but I wanted to try something while waiting for the 10 million sim

2.3 Full P_{orb}

When filtering population 2 with a max $p_{orb} = 300\text{days}$ we find 46 BH-Sol systems, all of these (except one!) evolve via the same CE channel. This suggests that there isn't a unique identity placed on BH-Sol systems with $p_{orb} \leq 3\text{days}$ ⁴, and thus it makes more sense to investigate the population as a whole. Note that there is one system which does *not* do RLO before CC1, and only reaches a solar-like mass after RLO2 (upon which it also fails), so I'm excluding it for now. Likely important to keep in mind though.

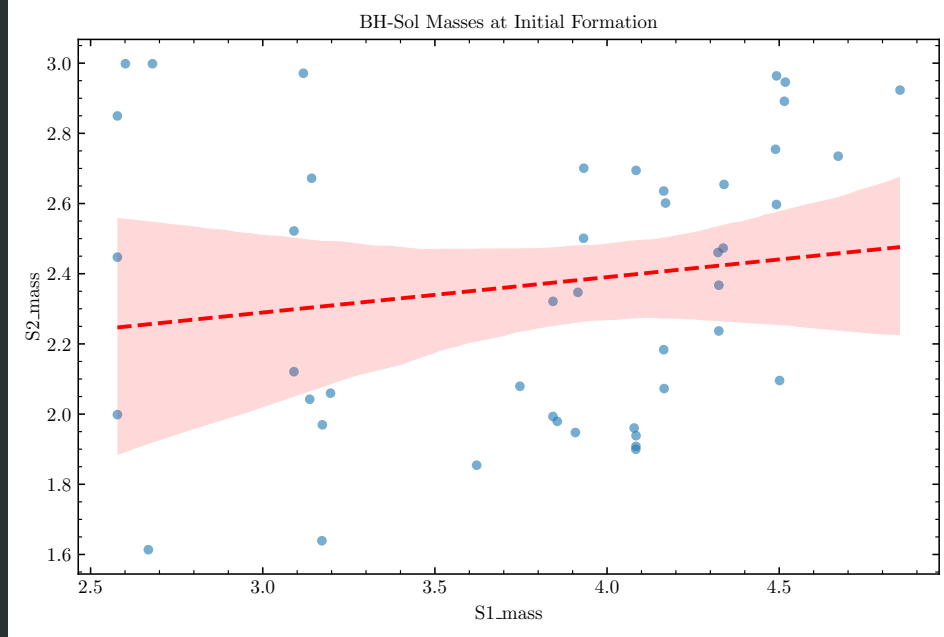


Figure 1: Masses at initial BH-Sol state. i.e., BH formation/S1 collapse

⁴This orbital period distribution is possibly due to natal kick velocities, and is suggested by the literature, but I could not find a strong link in the limited data.

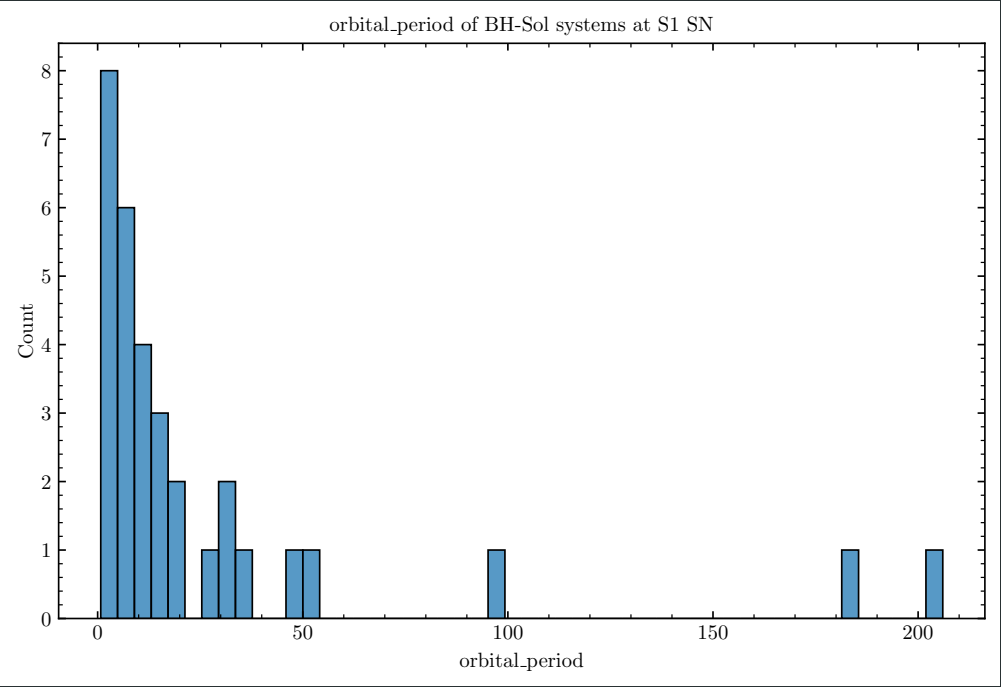


Figure 2: Orbital Period of BH-Sol Systems at init onset.

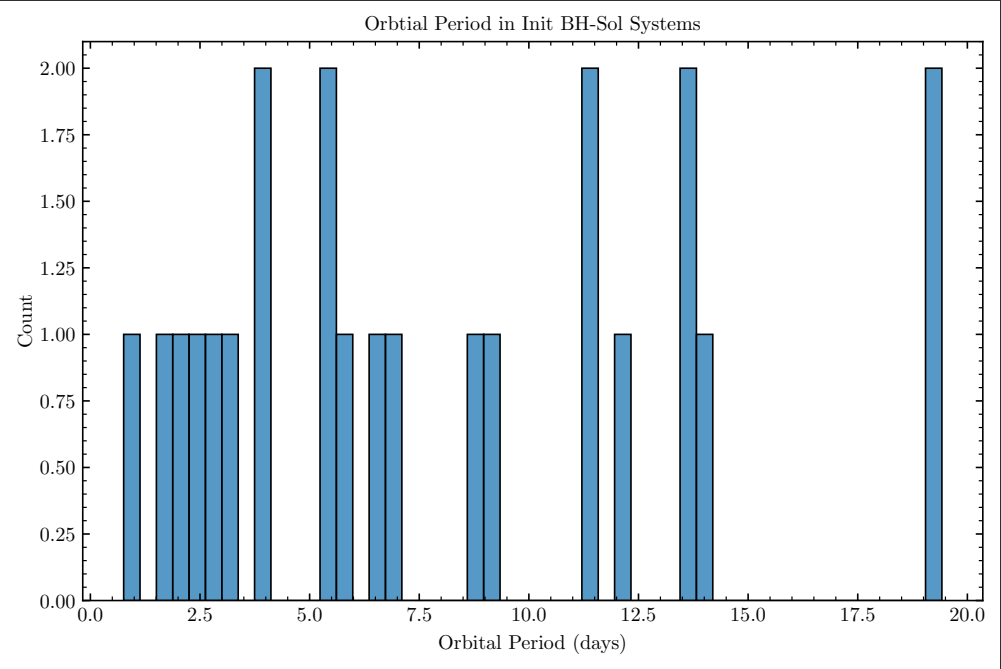


Figure 3: Crop of orbital Period of BH-Sol Systems at init onset.

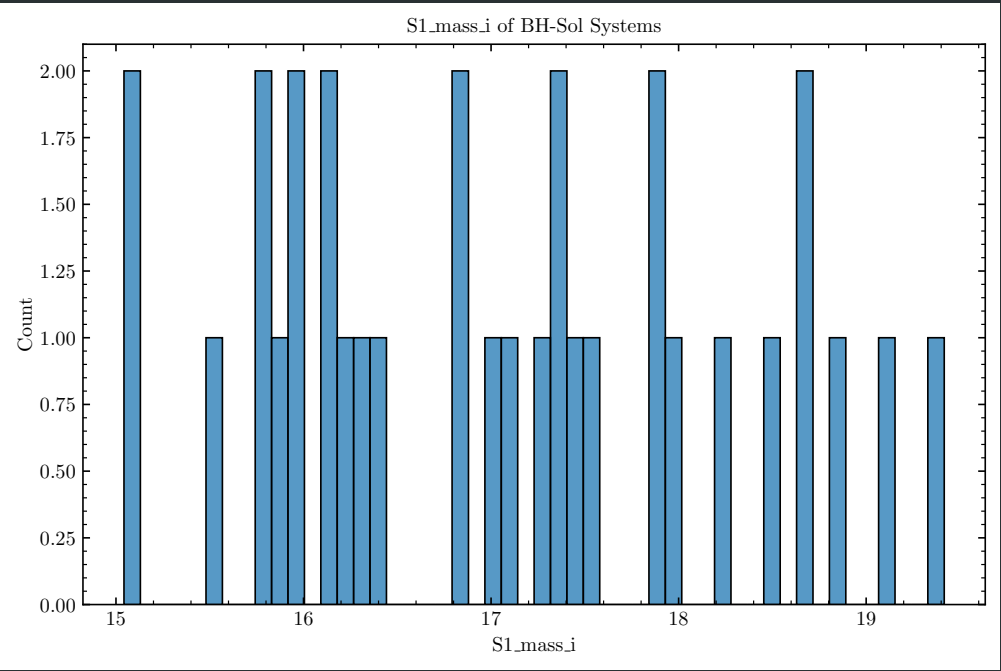


Figure 4: S1 mass initial distribution.

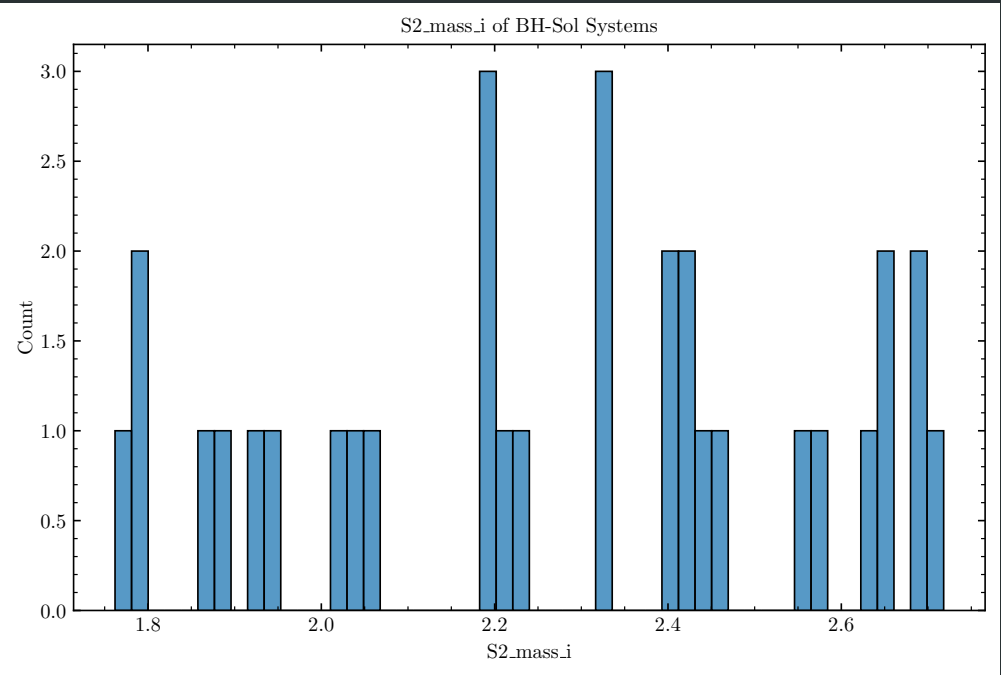


Figure 5: S2 mass initial distribution.

2.4 Strong Initial Masses Correlation

When plotting the individual masses of BH-Sol progenitors from Pop 2 there appeared to be a strong correlation between the two. We see this in figure 6.

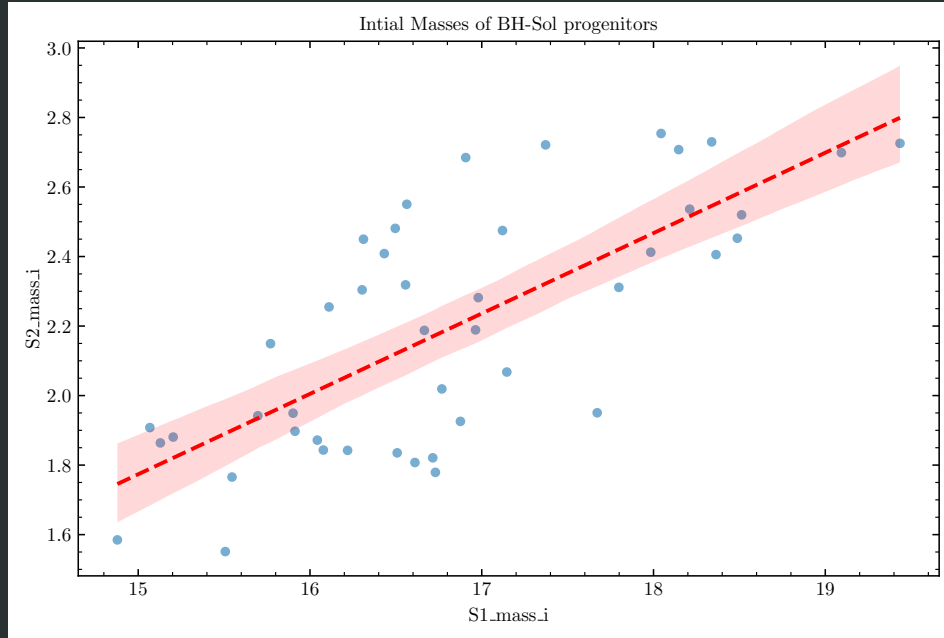


Figure 6: Initial ZAMS masses of simulation two, with mass ratios from Moe17PsandQs. *Note that two failed systems have been excluded*

Initially, I figured this was a biasing from the mass ratio scheme in the simulation being passed through, so I ran another grid with a `secondary_mass_scheme = 'flat_mass_ratio'`. While this would also introduce biasing, I was expecting it to be largely different enough from Moe, and thus could tell if it was simply an artifact or a real correlation. As we can see from figure 7 and 8, this correlation seems to be independent of the initial mass ratio used, suggesting some type of underlying phenomena⁵.

⁵This very likely could just be a survivability configuration of CE evolution

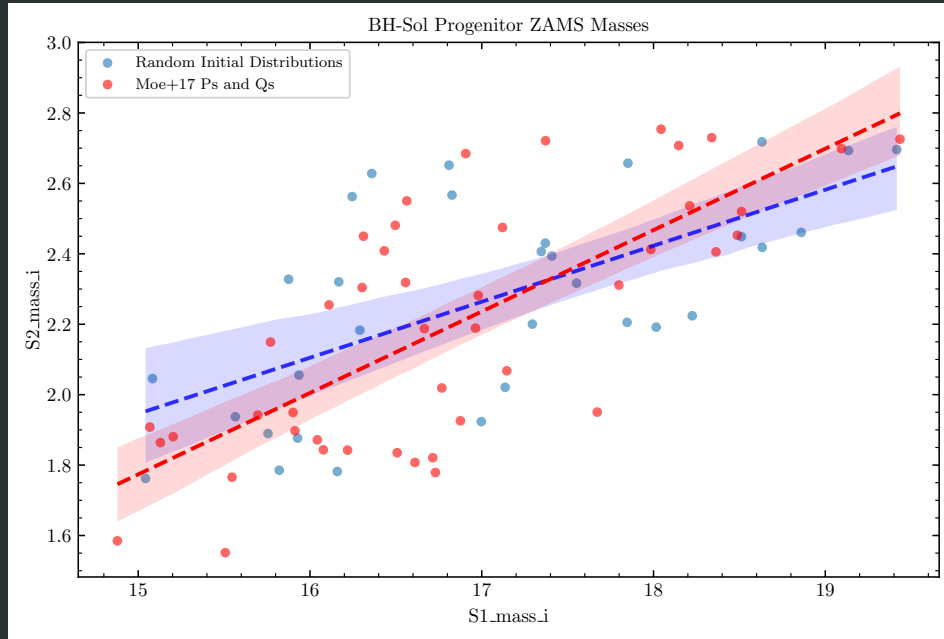


Figure 7: Initial BH-Sol ZAMS masses of simulation two (using Moe17PsandQs) and three (flat mass ratio). *Note that 3 failed systems have been excluded.*

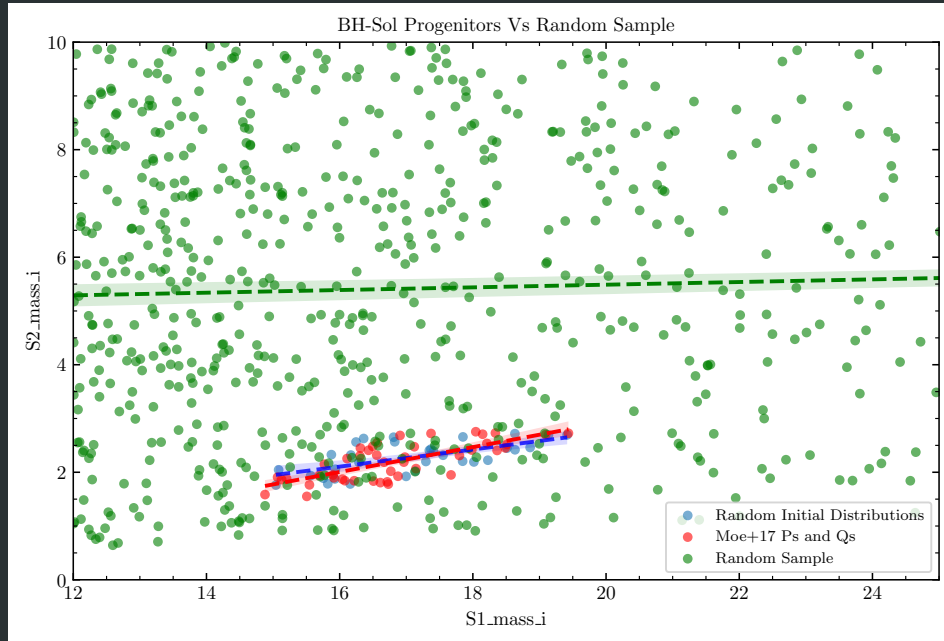


Figure 8: Initial BH-Sol ZAMS masses of simulation two and three, vs random selection of simulation 3. *Note that 3 failed BH-Sol systems have been excluded.*

3 To-Do

- Find out if it makes sense for the onset of unstable rlo to happen at $P_{orb} \approx 4000\text{days}$

4 Questions

- Tried running a grid, seems to have just completely failed from trying to reference itself before it was fully initialized. Entire log was filled with

```
Traceback (most recent call last):
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/popsyn/binarypopulation.py", 1
    binary.evolve()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    self.run_step()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
    next_step(self)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_merged.py", 1
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_isolated.py", 1
    super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py", 1
    self.update_after_evo(t, binary, primary, secondary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.py", 1
    current = history[-1]
                ~~~~~~
IndexError: index -1 is out of bounds for axis 0 with size 0
```

and from what I can tell, this happened to nearly all the simmed binaries

- Should we include an initial eccentricity distribution? I imagine it would impact formation with unstable RLO and CE.
- Does it make sense to still be filtering for $p_{orb} < 3\text{days}$? The simulations seem to be suggesting that *all* BH-Sol systems evolve through CE, and that the main thing effecting p_{orb} post CE is the kick strength of the BH. So, maybe we should be initially filtering for all BH-Sol, and if we want the formation rate of specifically low p_{orb} ones, just take the subpop.
- Cool if I present an assortment of general POSYDON things I've worked on at astrofest?